

Inhibitor Investigation

One-page teacher/presenter guide · Murray State University · Bryant Harrison

Purpose

Students use a four-measurement virtual assay bench to distinguish among three finite, declared *synthetic initial-rate patterns*: competitive, uncompetitive, and pure noncompetitive. They learn that a point can be exact and still be insufficient evidence.

Model. $v = V_{max} \cdot S / (a \cdot K_m + b \cdot S)$, with $V_{max} = 100$, $K_m = 2$, and inhibition factor 3. Baseline uses $a=b=1$; competitive $a=3, b=1$; uncompetitive $a=1, b=3$; pure noncompetitive $a=b=3$.

First-load / presenter flow

- Open `index.html` locally; the how-to opens automatically. Use ? to reopen it.
- Ask students which assay would be informative before they click. They may select 0, 1, 2, 4, 8, or 20 teaching units.
- At $S=K_m=2$, competitive and uncompetitive both predict 25.0 while pure noncompetitive predicts 16.7. Name this as an evidence boundary, not a failure.
- For this exact model, $S=1, 8$, or 20 each eliminates a candidate after that tie. Do not rank one as “best” by its curve gap; the model has no measurement-error or resolution rule.
- Students must explain before the mystery pattern is revealed. Open **Settings** for presentation mode and notes.

Discussion prompts

- What claim is supported by the plotted points? What claim is still too strong?
- Why does repeating the same concentration not add discrimination?
- Which assumptions would a real assay need to test before treating a fitted pattern as mechanism?

Expected synthetic predictions

Substrate S	Competitive	Uncompetitive	Pure noncompetitive
0	0.0	0.0	0.0
2 (=Km)	25.0	25.0	16.7
8	57.1	30.8	26.7
20	76.9	32.3	30.3

Scope and scientific caveats

This is an exact, noiseless, ideal initial-rate activity at fixed conditions. The tie at $S=K_m$ is deliberately constructed by the shared factor-3 set; it is not a general competitive/uncompetitive diagnostic. It is not an inhibitor fit, drug screen, or molecular-binding assignment.

The three patterns do not represent mixed, partial, allosteric, tight-binding, time-dependent, irreversible, multiple-substrate, or non-steady-state behavior. A real investigation would test a broader substrate/inhibitor matrix, controls, replicates, assay linearity, error, and model assumptions.

The activity intentionally lets students conclude **insufficient evidence**. It does not reward guessing the hidden generator, and it never claims that a curve pattern uniquely identifies a real mechanism.

Sources

Strelow J, Dewe W, Iversen PW, et al. Mechanism of Action Assays for Enzymes, NIH Assay Guidance Manual. Used for the inhibition-pattern and initial-rate assay context, including the value of measurements above K_m .

J. Cox, SciComm: Biochem. Used only as the reviewed Murray State course-fit reference; it does not endorse this activity or supply current scheduling information.